Exp No:15 Experiment to understand the data preprocessing in Data science

Description: Understand the importance of Data preprocessing in data science

Code:

import numpy as np

import pandas as pd

df = pd.read_csv('/content/pre-process_datasample.csv')

df

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No
9	France	37.0	67000.0	Yes

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 4 columns):

#	Column	Non-Null Count	Dtype
0	Country	10 non-null	object
1	Age	9 non-null	float64
2	Salary	9 non-null	float64
3	Purchased	10 non-null	object

dtypes: float64(2), object(2)

memory usage: 452.0+ bytes

df.Country.mode()

	Country
0	France

dtype: object

df.Country.mode()[0]

'France'

type(df.Country.mode())

df.Country.fillna(df.Country.mode()[0],inplace=True)

df.Age.fillna(df.Age.median(),inplace=True)

df.Salary.fillna(round(df.Salary.mean()),inplace=True)

df

pd.get_dummies(df.Country)

	France	Germany	Spain
0	True	False	False
1	False	False	True
2	False	True	False
3	False	False	True
4	False	True	False
5	True	False	False
6	False	False	True
7	True	False	False
8	False	True	False
9	True	False	False

updated_dataset=pd.concat([pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1)

updated_dataset

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	True	False	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	40.0	63778.0	Yes
5	True	False	False	35.0	58000.0	Yes
6	False	False	True	38.0	52000.0	No
7	True	False	False	48.0	79000.0	Yes
8	False	True	False	50.0	83000.0	No
9	True	False	False	37.0	67000.0	Yes

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 4 columns):

#	Column	Non-Null Count	Dtype
0	Country	10 non-null	object
1	Age	10 non-null	float64
2	Salary	10 non-null	float64
3	Purchased	10 non-null	object

dtypes: float64(2), object(2)

memory usage: 452.0+ bytes

Result:

Understood the importance of Data preprocessing in data science